

Online Appendix: Exposure to the COVID-19 Stock Market Crash and its Effect on Household Expectations

Tobin Hanspal* Annika Weber[†] Johannes Wohlfart[‡]

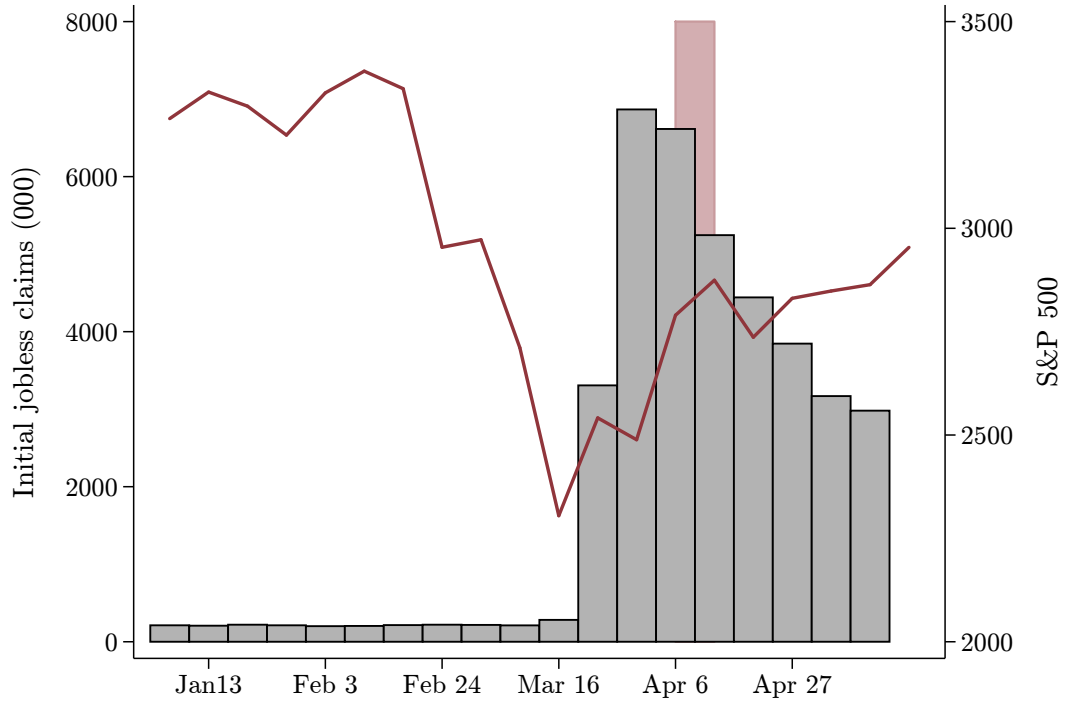
*Tobin Hanspal, Department of Finance, Accounting and Statistics, WU Vienna University of Economics and Business, e-mail: tobin.hanspal@wu.ac.at

[†]Annika Weber, Department of Finance, Goethe University Frankfurt, e-mail: annika.weber@hof.uni-frankfurt.de

[‡]Johannes Wohlfart, Department of Economics and CEBI, University of Copenhagen, CESifo, e-mail: johannes.wohlfart@econ.ku.dk

A Additional figures

Figure A1: US stock market and number of initial jobless claims around the survey period



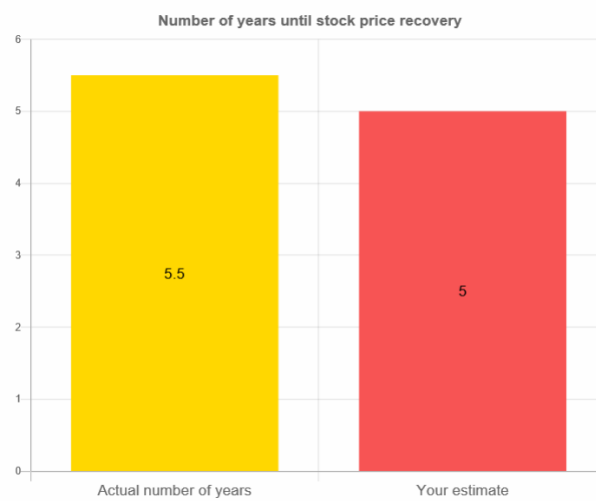
Notes: This figure displays the number of initial jobless claims (in thousands, left axis) and the development of the S&P500 stock market index (index points, right axis) over the first 19 weeks in 2020, on a weekly basis. The April 6-13 survey period is highlighted in light red.

Figure A2: Information Treatment *FinCrisisInfo*

You estimated that stock prices recovered to their pre-crisis level 5 years after the beginning of their drop in October 2007.

We now provide you with information on the actual development of stock prices during the Financial Crisis that started in 2007.

Starting from the beginning of the drop in stock prices in October 2007, it took **about 5 ½ years** until stock prices recovered to their pre-crisis level in March 2013.



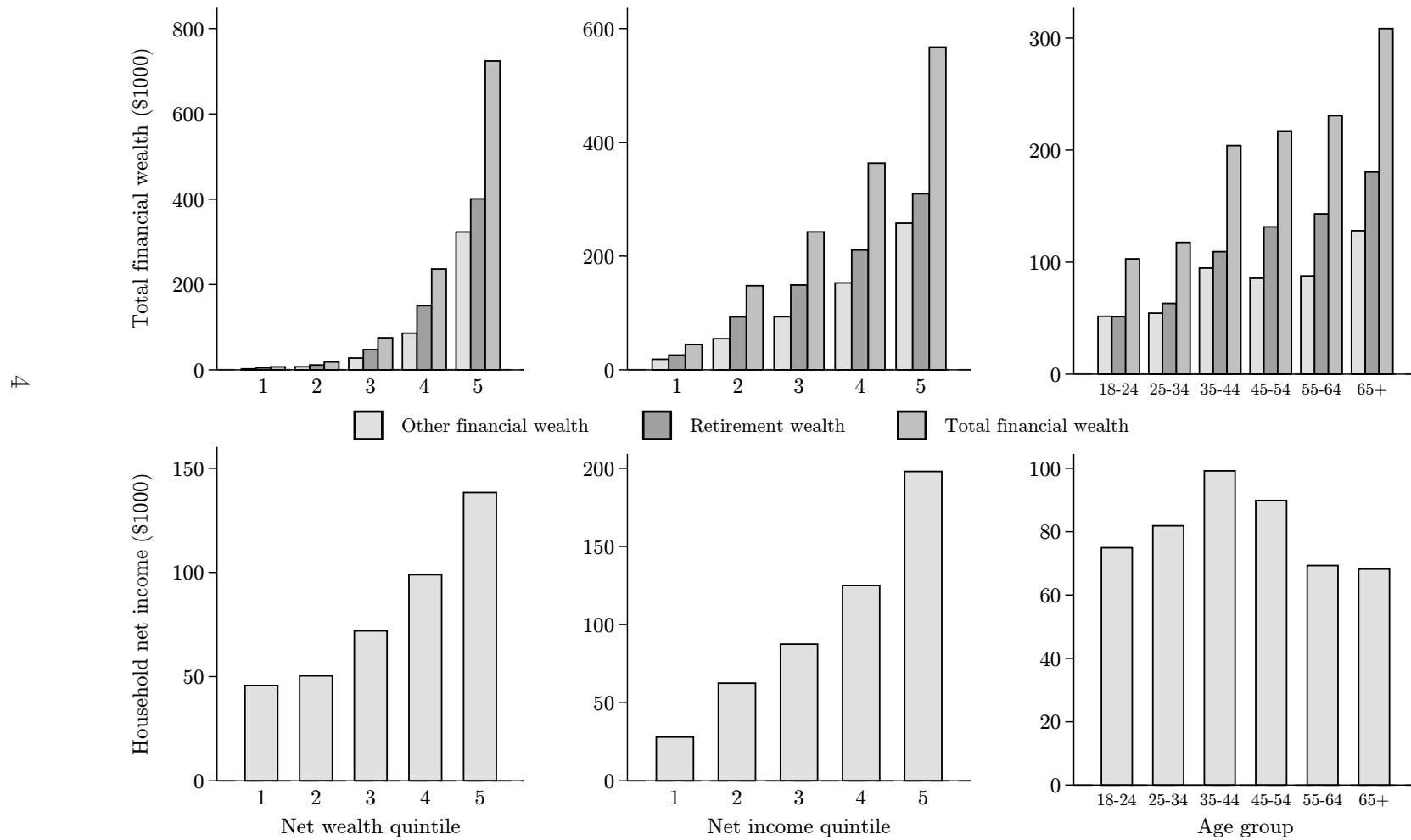
Source: The number is based on the time that the stock market index S&P500 needed to recover to its pre-crisis level.

Note: You will be able to go to the next slide after you have spent a few seconds on this slide.



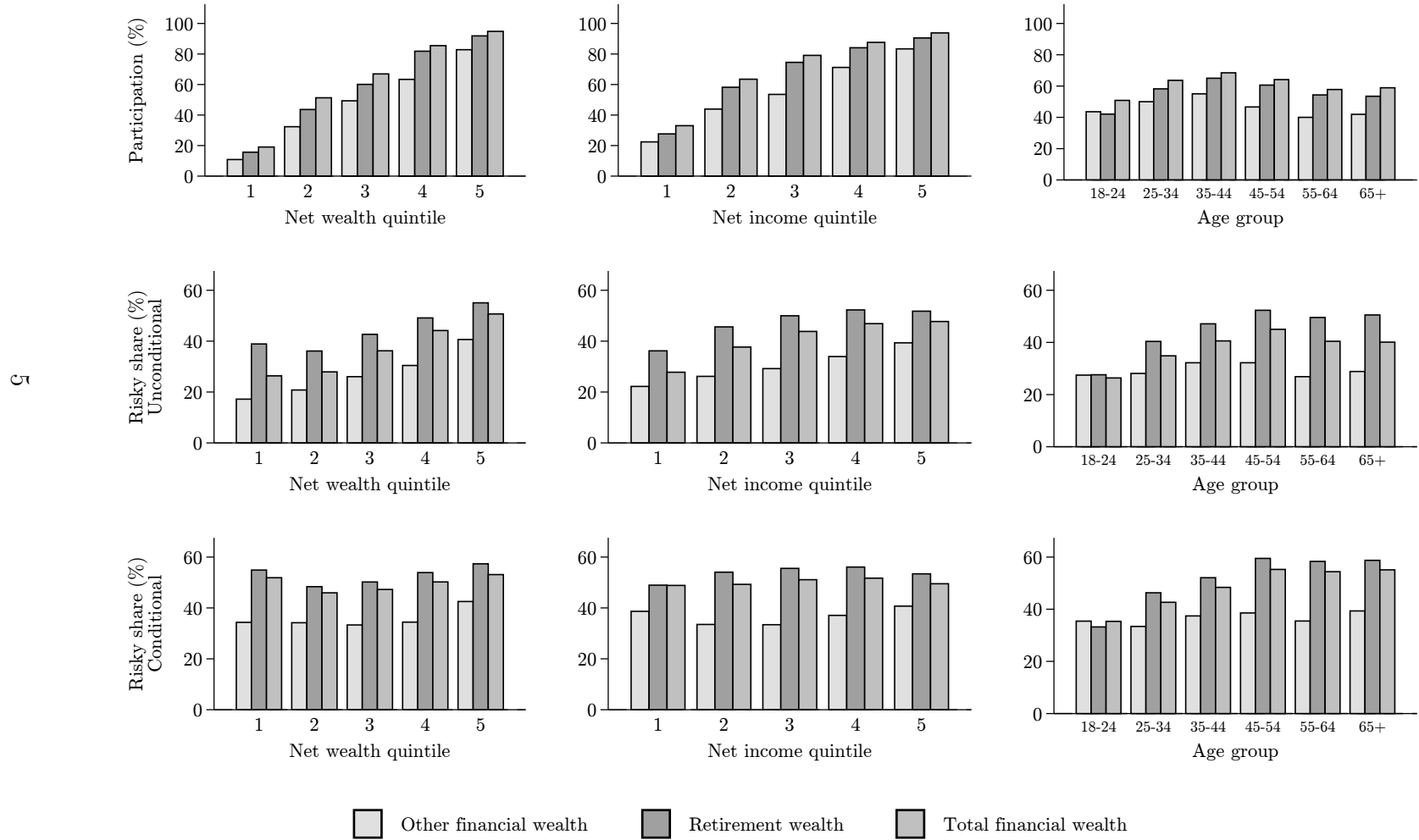
Notes: This figure illustrates the information treatment screen, providing an example of the *FinCrisisInfo* treatment arm. The information treatment includes a dynamic figure contrasting the respondent's prior belief (in dark orange, on the right) with the actual number of years it took for the US stock market to recover to its levels before the 2007-2009 Financial Crisis (in yellow, on the left). Recovery durations for the three different information treatments *FinCrisisInfo*, *BlackMondayInfo* and *DotComInfo* are calculated based on monthly time series data of the S&P500.

Figure A3: Financial assets and incomes across groups



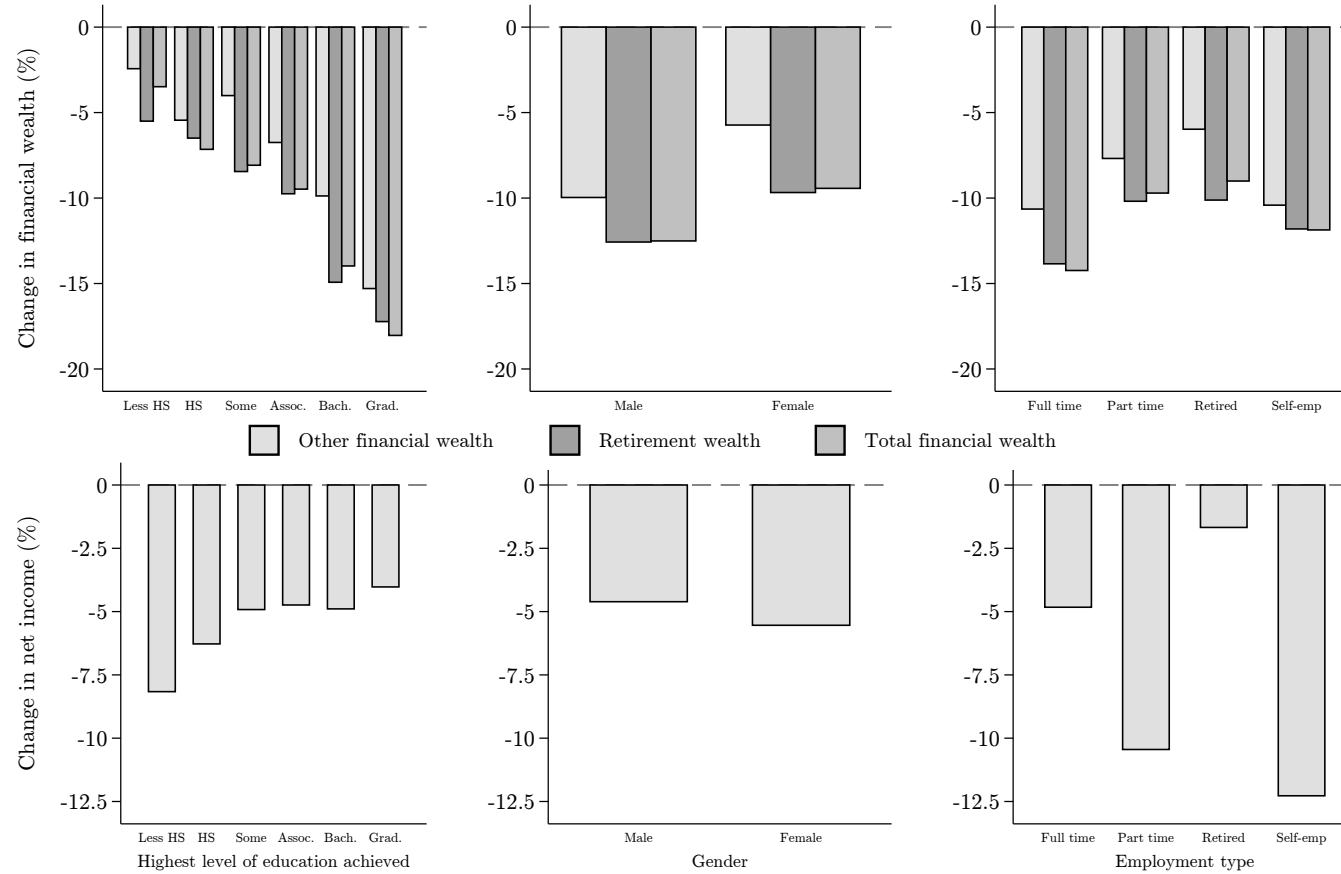
Notes: This figure displays the average value of financial assets (top row) and gross household income during the first quarter of 2020 (bottom row), by quintile of the pre-crisis net worth distribution (left column), by quintile of the pre-crisis net income distribution (middle column) and by age group (right column). Values for financial assets are displayed separately for financial assets outside of retirement accounts, for financial assets in retirement accounts, and for the combined value of all financial assets. The sample is the full sample without missings in the relevant survey questions.

Figure A4: Participation and stock share of financial wealth across groups



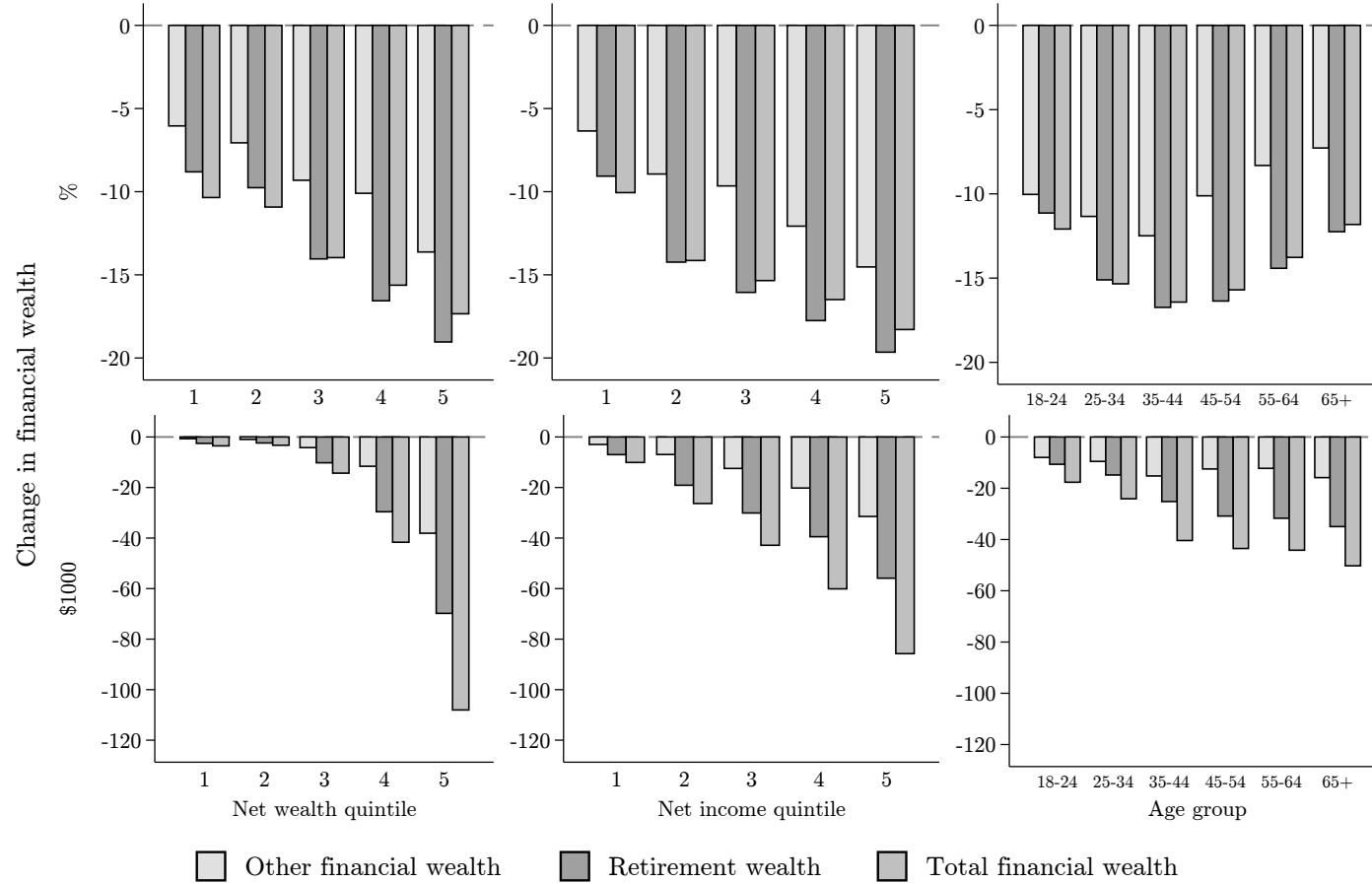
Notes: This figure displays participation, and the share of financial wealth held, in stocks or mutual funds by quintiles of pre-crisis net wealth (left column) of pre-crisis net income, (middle), and by age group (right), respectively. The top row plots the rate of participation in stocks and stock mutual funds in the sample. The middle row plots the unconditional equity share, and the bottom plots the conditional equity share including only respondents that report positive holdings of stocks or stock mutual funds as of January 2020. Equity shares are displayed separately for financial assets outside retirement accounts, for financial assets in retirement accounts, and for the combined value of financial assets. The sample includes all respondents without missings in the relevant survey questions.

Figure A5: Income and wealth shocks across groups



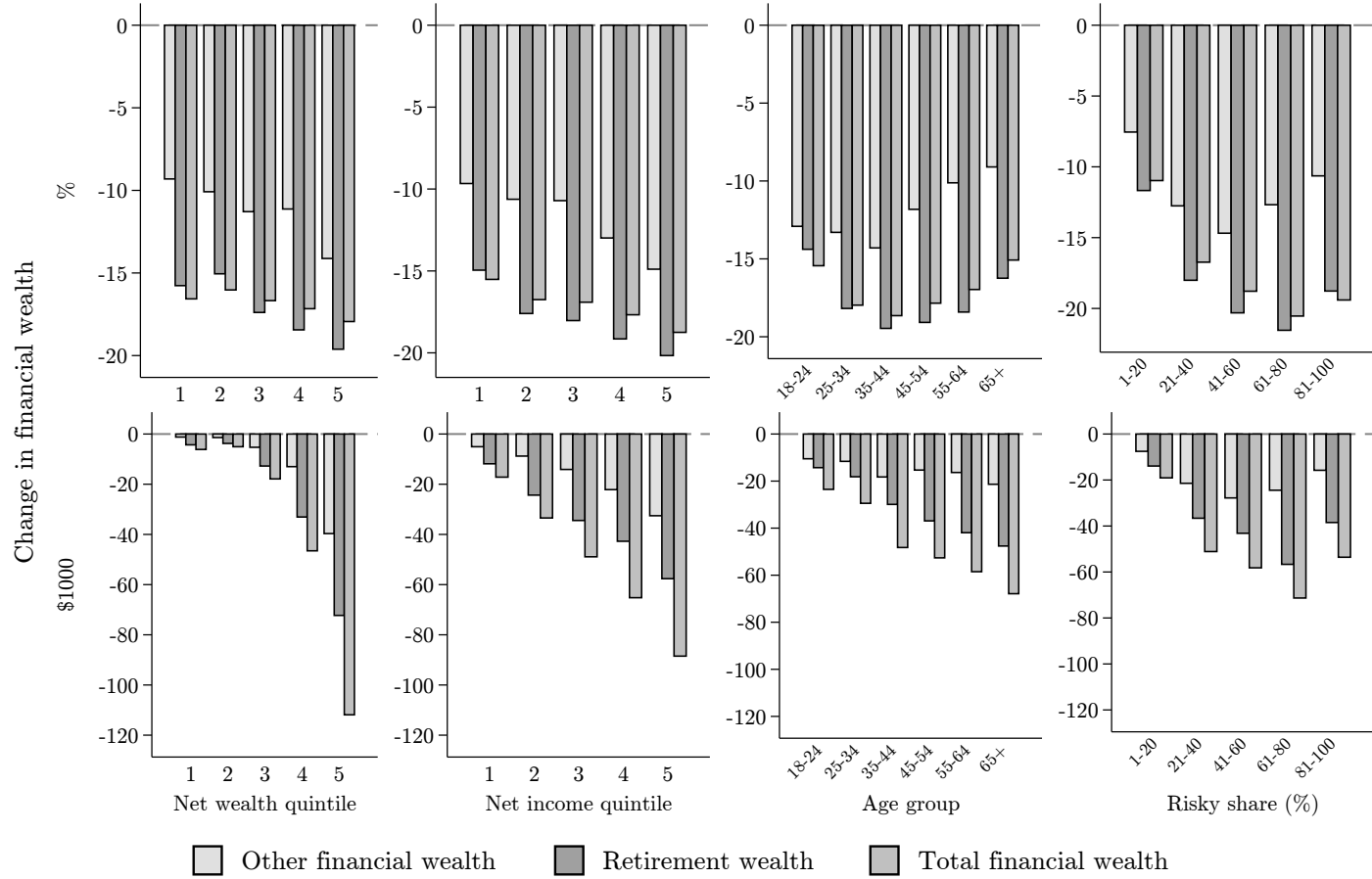
Notes: This figure displays the change in the value of financial assets due to the February/March 2020 stock market drop until the survey date in percentage terms as amounts in USD (top panel) and unexpected changes in net household incomes during the first quarter of 2020 in percentages and USD amounts (bottom panel) by highest level of education achieved (left column), gender (middle column), and pre-crisis employment type (right column). Changes in the value of financial assets are displayed separately for financial assets outside of retirement accounts, for financial assets in retirement accounts, and for the combined value of all financial assets. Changes in value of financial assets are net capital losses for the majority of respondents, and net capital gains for a small fraction of respondents. We report reported shocks to income and financial wealth at the 2nd and 98th percentiles. The sample is the full sample without missings in the relevant survey questions. Note that the average percent reduction in overall financial wealth can be larger than both average percent reductions for the individual components. This is due to the fact that we coded those with no wealth in a given category as having experienced a shock of zero percent in that category. These cases occur particularly in groups with lower wealth holdings.

Figure A6: Conditional wealth shocks across groups (Financial wealth > 0)



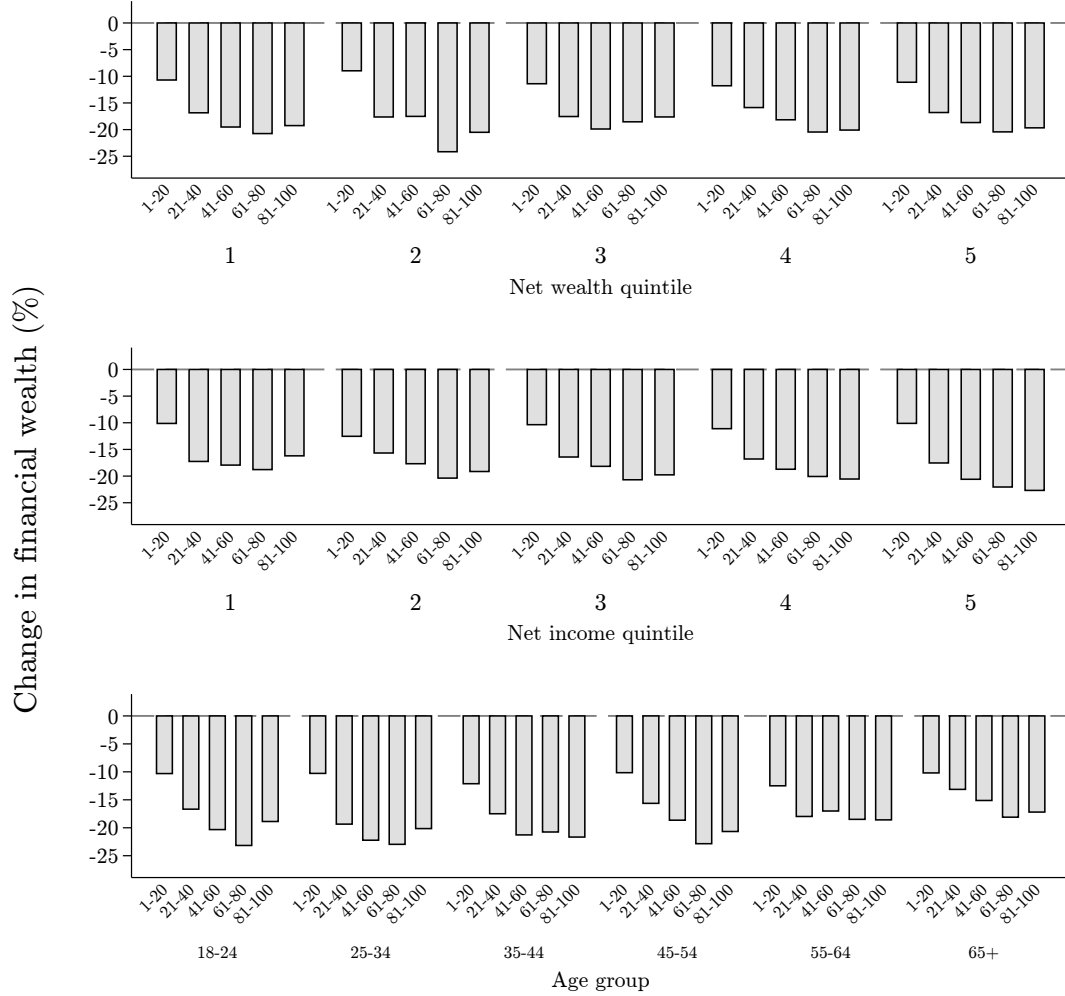
Notes: This figure displays the change in the value of financial assets due to the February/March 2020 stock market drop until the survey date in percentage terms and as amounts in USD, by quintile of the pre-crisis net worth distribution (left column), by quintile of the pre-crisis net income distribution (middle column), and by age group (right column). Changes in the value of financial assets are displayed separately for financial assets outside of retirement accounts, for financial assets in retirement accounts, and for the combined value of all financial assets. The values are conditional on positive financial wealth in January 2020. Changes in the value of financial assets are net capital losses for the majority of respondents, and net capital gains for a small fraction of respondents. We trim reported shocks to financial wealth at the 2nd and 98th percentiles. The sample is the full sample without missings in the relevant survey questions. Note that the average percent reduction in overall financial wealth can be larger than both average percent reductions for the individual components. This is due to the fact that we coded those with no wealth in a given category as having experienced a shock of zero percent in that category. These cases occur particularly in groups with lower wealth holdings.

Figure A7: Conditional wealth shocks across groups (Risky share > 0)



Notes: This figure displays the change in the value of financial assets due to the February/March 2020 stock market drop until the survey date in percentage terms and as amounts in USD, by quintile of the pre-crisis net worth distribution (left column), by quintile of the pre-crisis net income distribution (second column), by age group (third column), and by bin of the equity share in total financial wealth. Changes in the value of financial assets are displayed separately for financial assets outside of retirement accounts, for financial assets in retirement accounts, and for the combined value of financial assets. The values are conditional on having positive equity investments in January 2020. Changes in the value of financial assets are net capital losses for the majority of respondents, and net capital gains for a small fraction of respondents. We trim reported shocks to financial wealth at the 2nd and 98th percentiles. The sample is the full sample without missings in the relevant survey questions. Note that the average percent reduction in overall financial wealth can be larger than the average percent reductions for both individual components. This is due to the fact that we coded those with no wealth in a given category as having experienced a shock of zero percent in that category. This occurs particularly in groups with lower wealth holdings.

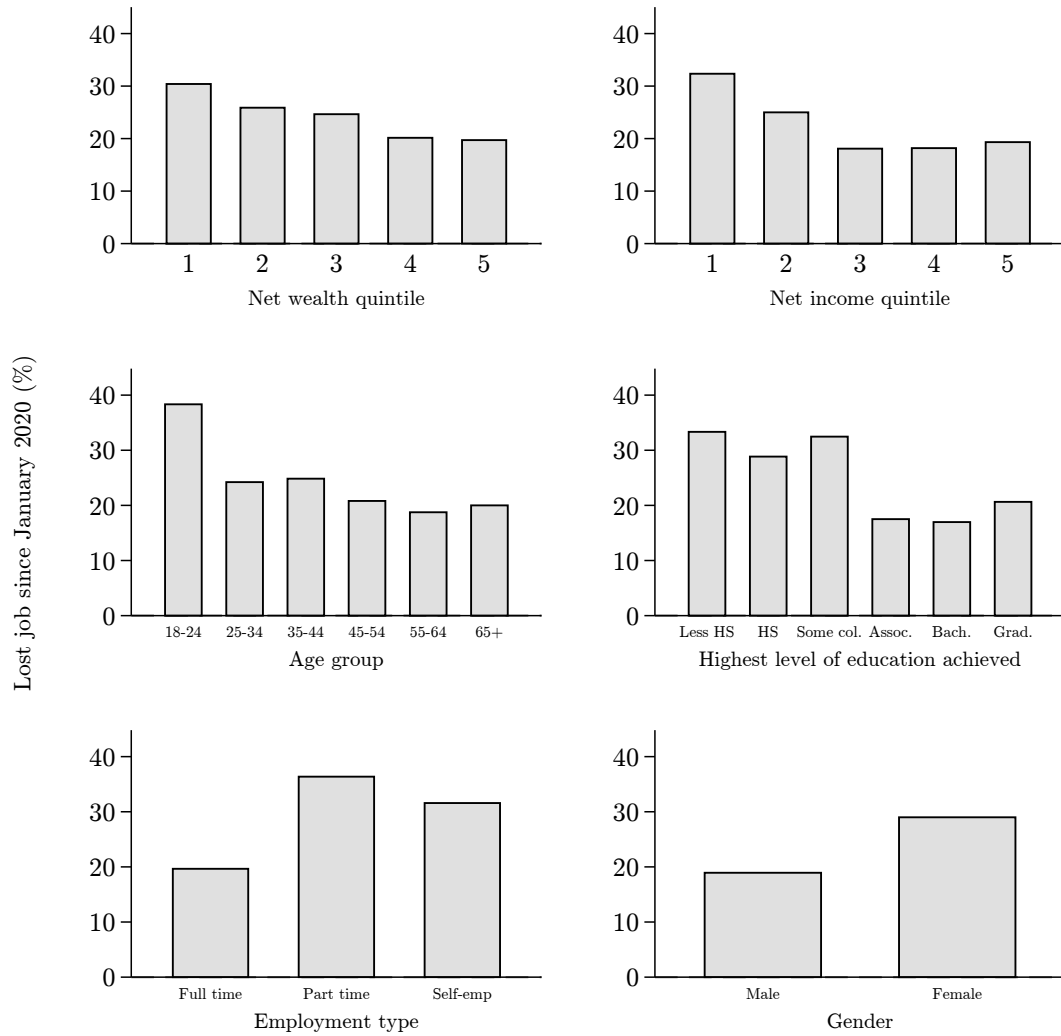
Figure A8: Wealth shocks across groups by risky share



January 2020 risky share allocation

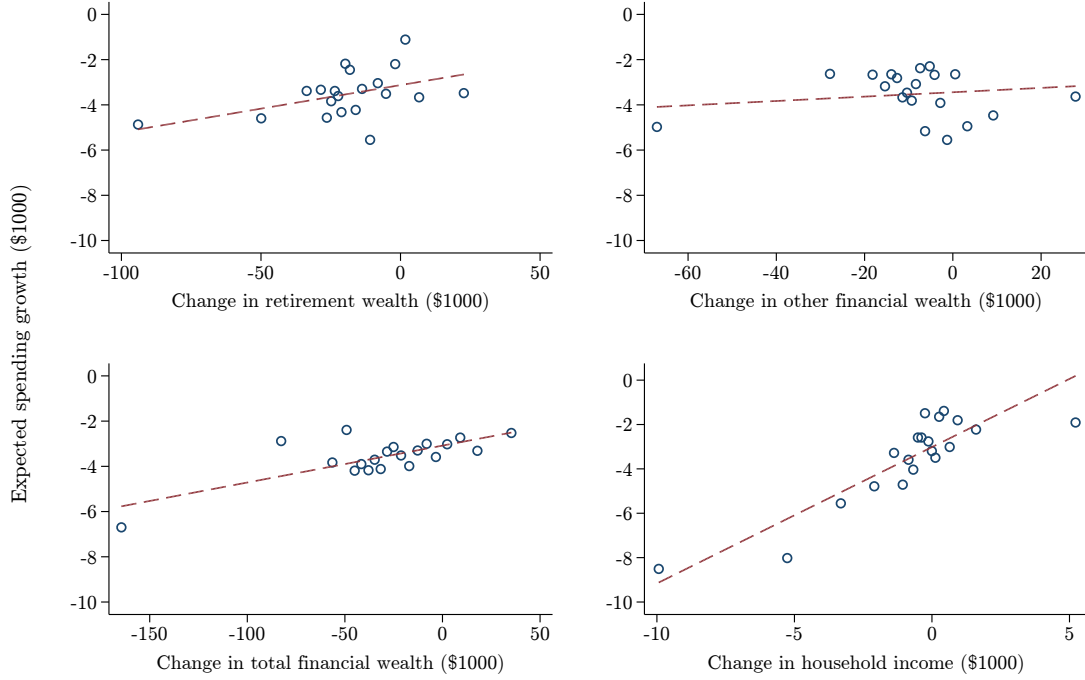
Notes: This figure displays the change in the value of financial assets due to the February/March 2020 stock market drop until the survey date in percentage terms by equity portfolio share bin, separately by quintile of the pre-crisis net worth distribution (top row), by quintile of the pre-crisis net income distribution (middle row), by age group (bottom row). Changes in the value of financial assets are for the combined value of financial assets inside and outside of retirement accounts. The values are conditional on positive equity investments in January 2020. Changes in the value of financial assets are net capital losses for the majority of respondents, and net capital gains for a small fraction of respondents. We trim reported shocks to financial wealth at the 2nd and 98th percentiles. The sample is the full sample without missings in the relevant survey questions.

Figure A9: Job losses across groups



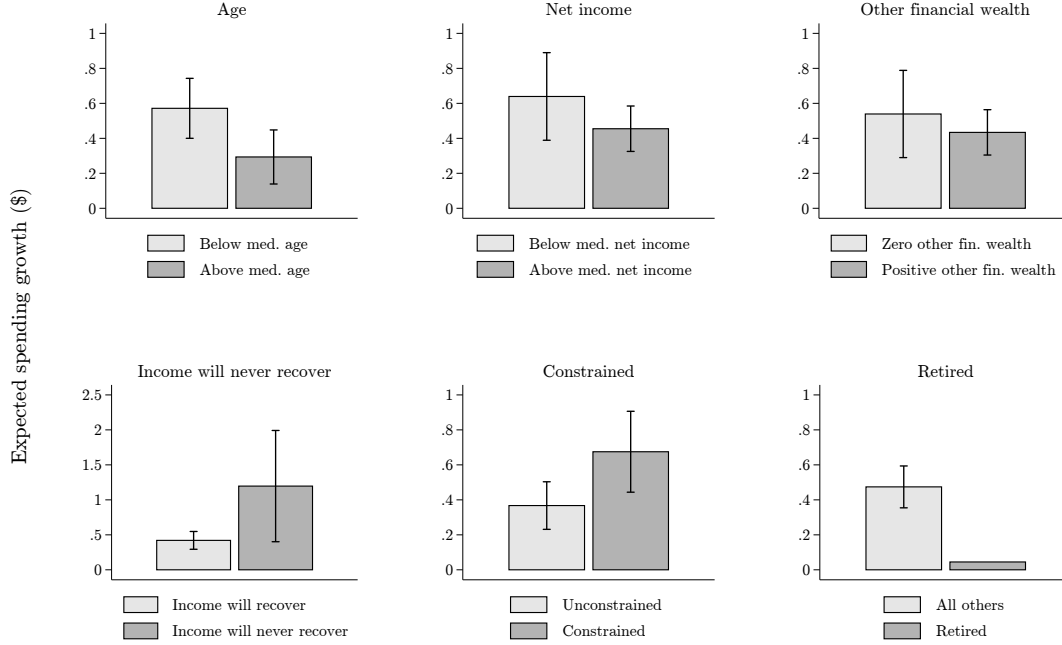
Notes: This figure displays the percentage of respondents who lost their job since January 2020 by pre-crisis net wealth quintile (top left), pre-crisis net income quintile (top right), age (middle left), education (middle right), pre-crisis employment type (bottom left), and gender (bottom right). The question is presented only to respondents who report to have been employed as of January 2020. The sample is the full sample without missings in the relevant survey questions.

Figure A10: Effects of wealth and income shocks on expected spending



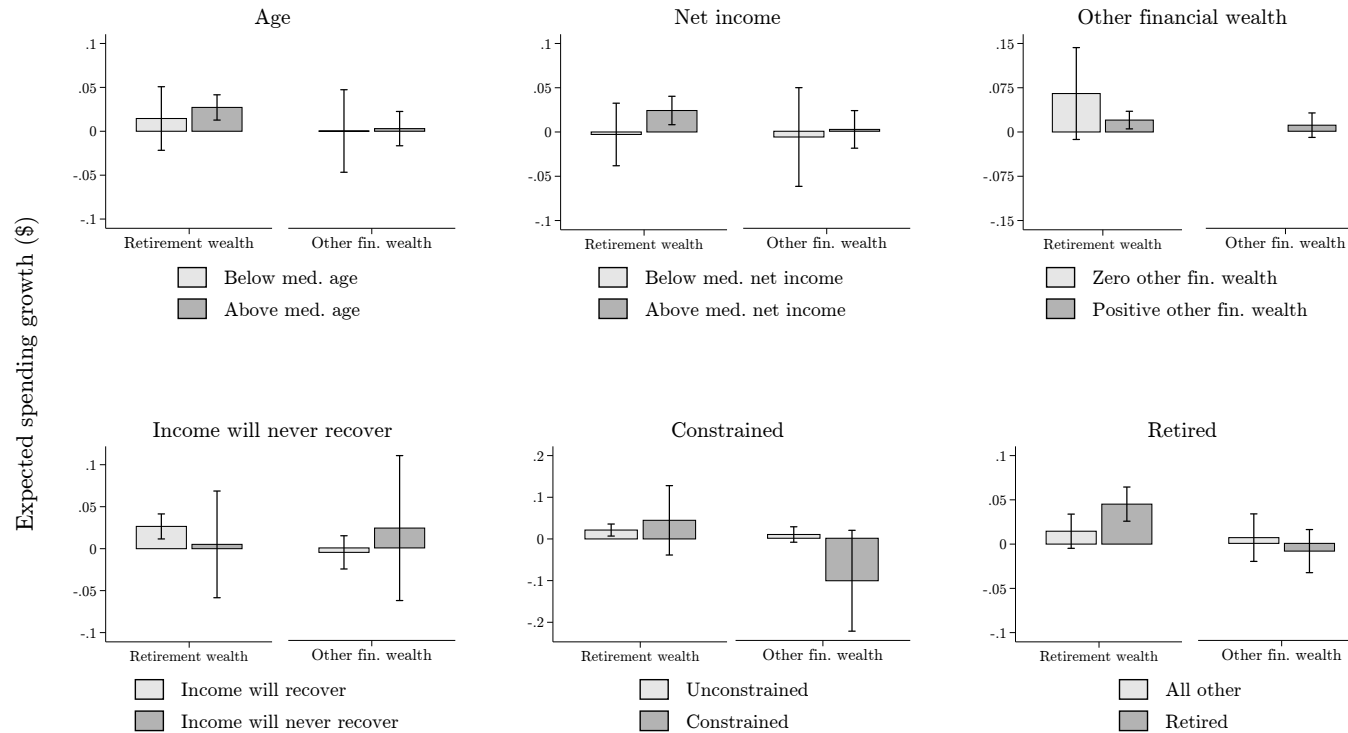
Notes: This figure shows binned scatter plots of the association of shocks to the respondent's household financial wealth and net income with expected growth of total nominal household spending in 2020 compared to 2019. The plots in the top row and in the bottom right are based on specification 2 shown in Table ??, which jointly includes shocks to income, retirement financial wealth and other financial wealth. The plot in the bottom left is based on a similar specification replacing the shocks to financial wealth in retirement accounts and in non-retirement accounts with the shock to overall financial wealth. The outcome is expected household spending growth in dollars, trimmed at the 2nd and 98th percentiles. Dollar changes are constructed from survey questions for retirement and other financial wealth and for income (assuming that the respondent expected a quarter of her 2019 income for the first quarter of 2020), and for spending from the survey question on percent changes and estimates of the level of spending of different groups in 2019 from the CEX. All specifications are based on respondents in the four control arms, who have not received any information. All specifications control for gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in total financial assets, investment experience, Census region, survey date, and the survey arm.

Figure A11: Effects of income shocks on expected spending: Heterogeneity



Notes: This figure shows heterogeneity in the effect of income shocks on expected growth of total nominal household spending in 2020 compared to 2019 across groups. The plots are based on the 2SLS specification shown in Table ?? column 3, where the respondent's expected dollar change in household income from 2019 to 2020 is instrumented with the unexpected shock to household income in the first quarter, estimated for different subsamples. The different panels show median splits (below and equal to median vs. strictly above median) according to age (top left) and pre-crisis household net income (top middle), by an indicator (0 vs. 1) for holding other (non-retirement) financial wealth in January 2020 (top right), by an indicator for believing household income will never recover (bottom left), by an indicator for the respondent's household facing credit constraints (bottom middle), and being retired (bottom right). The outcome is expected household spending growth in dollars, trimmed at the 2nd and 98th percentiles. Dollar changes are constructed from survey questions for retirement and other financial wealth and for income (assuming that the respondent expected a quarter of her 2019 income in the first quarter of 2020), and for spending from the survey question on percent changes and estimates of the level of spending of different groups in 2019 from the CEX. In the bottom right panel we do not include confidence bands as the interval is large and insignificant. For all others we include 90% confidence intervals. All specifications are based on respondents in the four control arms, who have not received any information. All specifications control for gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in total financial assets, investment experience, Census region, survey date, and the survey arm.

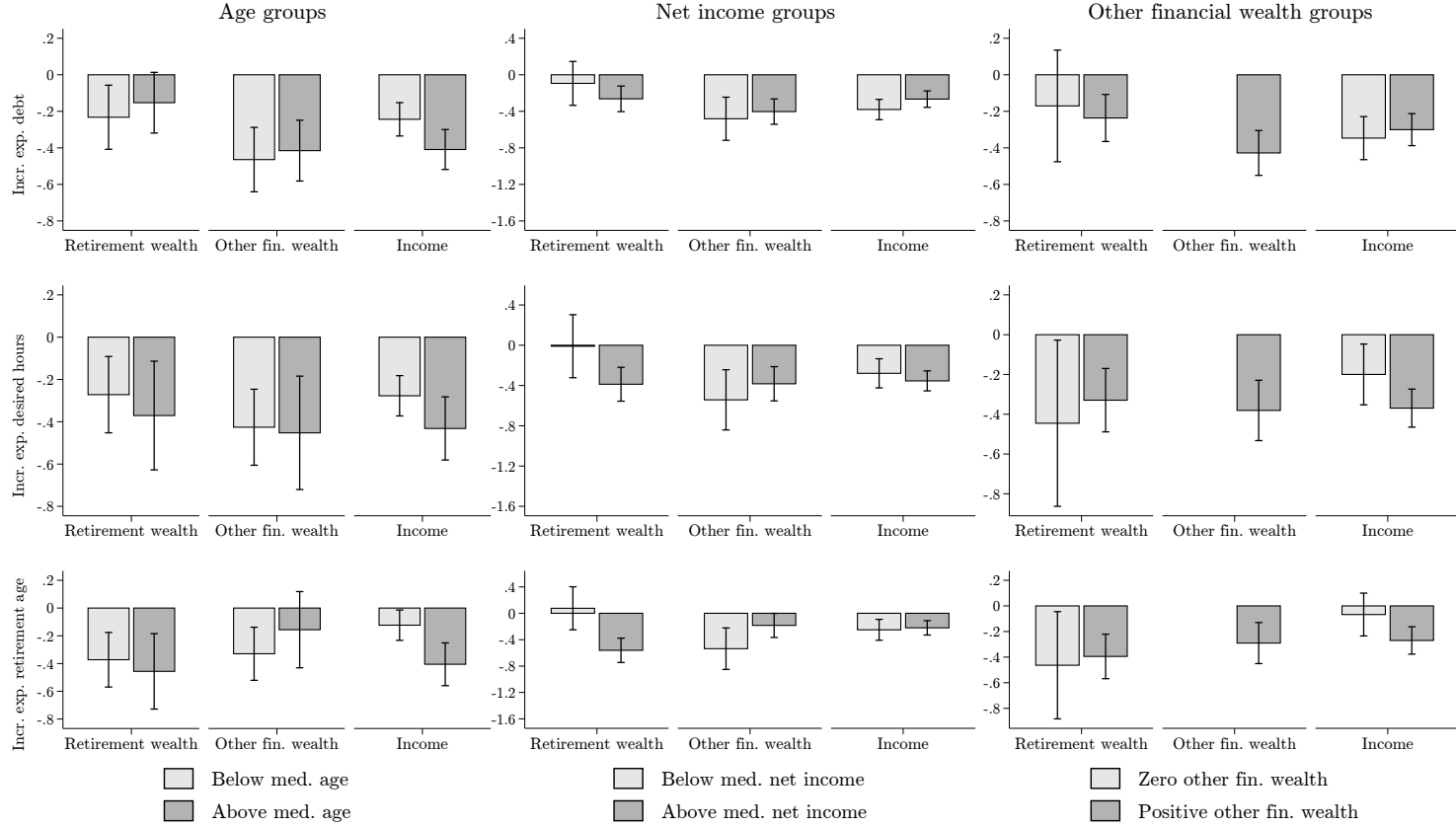
Figure A12: Effects of wealth shocks on expected spending: Heterogeneity



Notes: This figure shows heterogeneity in the association of shocks to the respondent's household retirement financial wealth and non-retirement financial wealth and the expected growth of total nominal household spending in 2020 compared to 2019. The plots for wealth shocks are based on the reduced-form specifications controlling for the first quarter-income shock shown in Table ?? column 2, estimated for different subsamples. The outcome is expected household spending growth in dollars, trimmed at the 2nd and 98th percentiles. Dollar changes are constructed from survey questions for retirement and other financial wealth and for spending from the survey question on percent changes and estimates of the level of spending of different groups in 2019 from the CEX. We plot coefficients on changes in respondent's household retirement financial wealth and other financial wealth by median splits (below and equal to median vs. strictly above median) according to age (top left) and pre-crisis household net income (top middle), by an indicator (0 vs. 1) for holding other (non-retirement) financial wealth in January 2020 (top right), by an indicator for believing individual income will never recover (bottom left), by an indicator for the respondent's household facing credit constraints (bottom middle), and by being retired (bottom right). 90% confidence intervals are displayed in all figures. All specifications are based on respondents in the control arms, who have not received any information. All specifications control for gender, age, employment status, being the main earner, being financial decision maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, risky portfolio share, investment experience, Census region, survey date, and the survey arm.

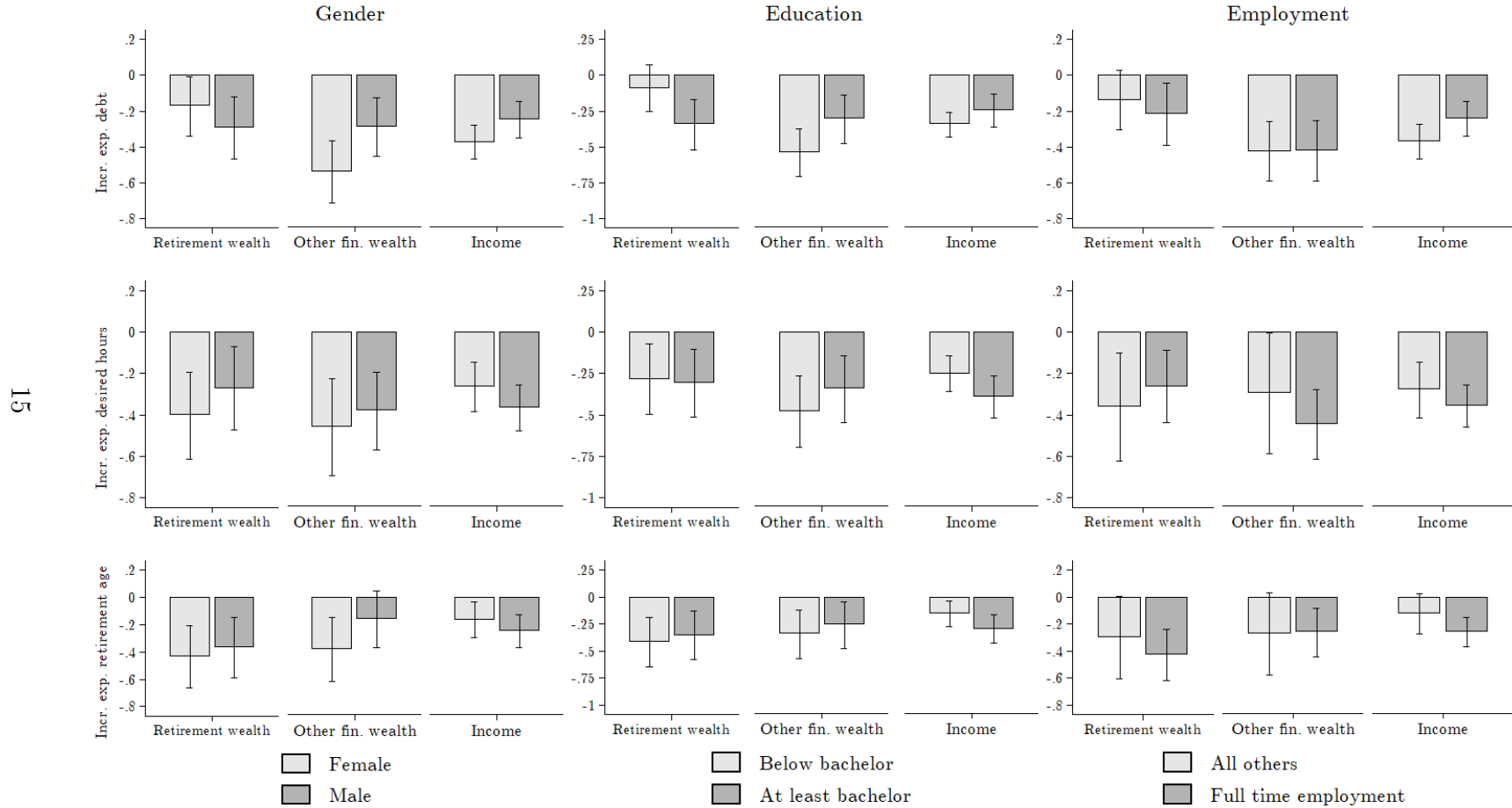
Figure A13: Effects of wealth and income shocks on economic plans: Heterogeneity I

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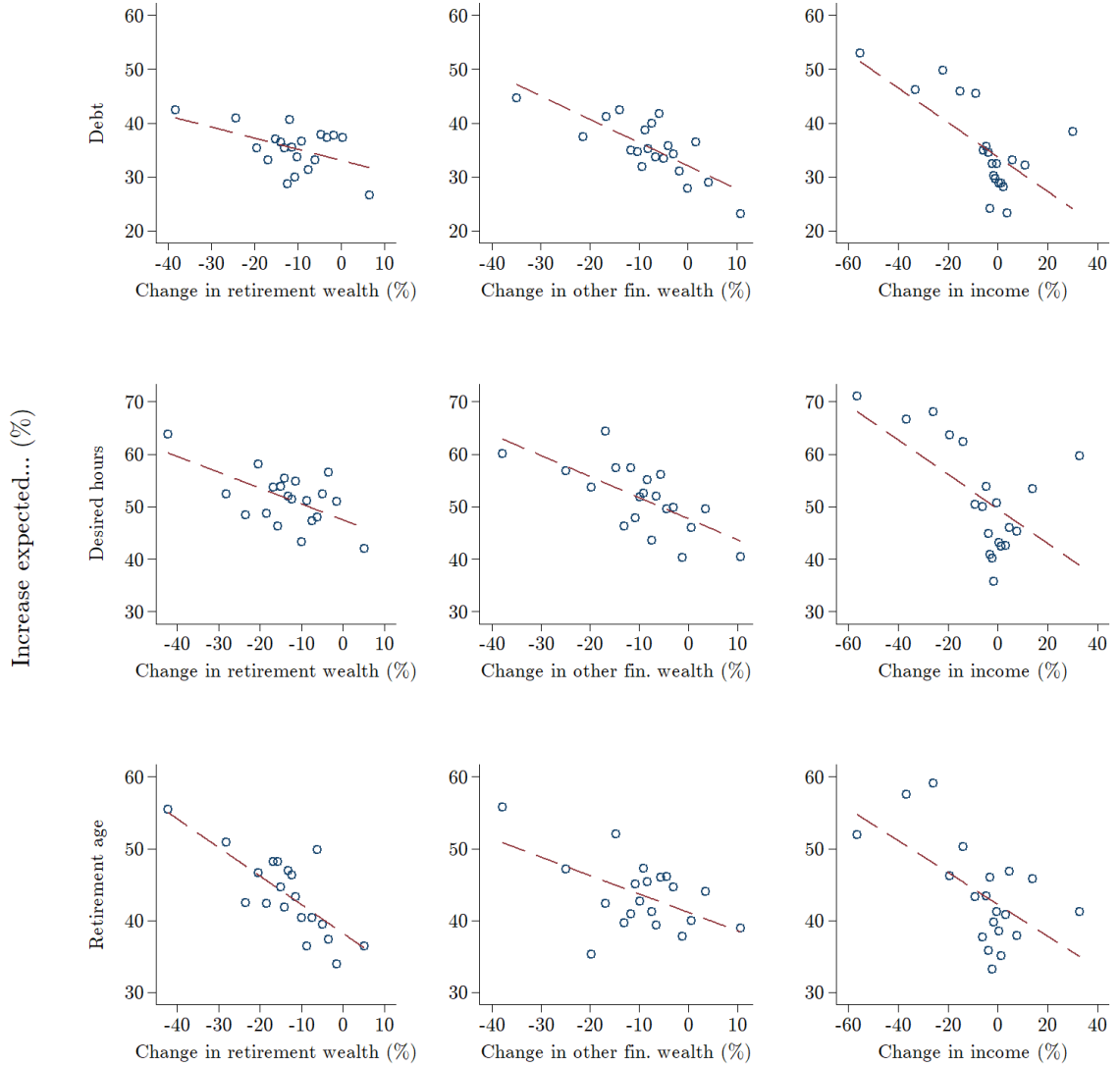
Notes: This figure shows heterogeneity in the effect of shocks to the respondent's household financial wealth and net income on expected economic decisions. We plot the coefficients from specifications regressing indicators for whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (top row), expected desired working hours over the next years (middle row) and expected retirement age (bottom row), all coded as 100, on changes to a household's retirement financial wealth, other financial wealth, and net household income. We plot these coefficients across columns by median splits (below and equal to median vs. strictly above median) according to age (left column) and pre-crisis household net income (middle), and by an indicator (0 or 1) for holding other (non-retirement) financial wealth in January 2020 (right). 90% confidence intervals are displayed in all figures. All specifications are based on respondents in the four control arms, who have not received any information. All specifications control for gender, age, employment status, being the household's main earner, being the household's financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of estate wealth, and of debt, borrowing constraints, stock market participation, the share of equity in total financial assets, investment experience, Census region, survey date, and the survey arm.

Figure A14: Effects of wealth and income shocks on economic plans: Heterogeneity II



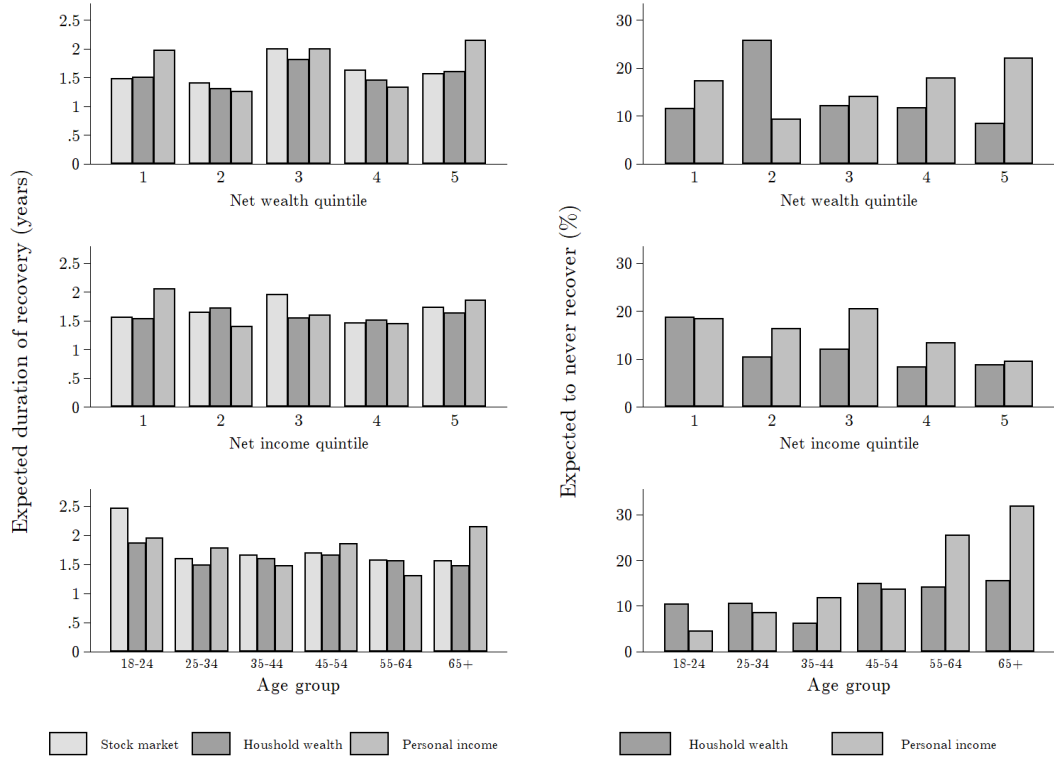
Notes: This figure shows heterogeneity in the effect of shocks to the respondent's household financial wealth and net income on expected economic decisions. We plot the coefficients from specifications regressing indicators for whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (top row), expected desired working hours over the next years (middle row) and expected retirement age (bottom row), all coded 0 or 100, on changes to a household's retirement financial wealth, other financial wealth, and net household income. We plot these coefficients across columns for indicators for gender (left column), education of at least a bachelors degree (middle), and being full-time employed pre-crisis (right). 90% confidence intervals are displayed on all figures. All specifications are based on respondents in the four control arms, who have not received any information. All specifications control for gender, age, employment status, being the household's main earner, being the household's financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in the household's financial assets, investment experience, Census region, survey date, and the survey arm.

Figure A15: Effects of wealth and income shocks on economic plans



Notes: This figure shows binned scatter plots of the association of shocks to the respondent's household financial wealth and net income with expected economic decisions. The outcomes are a dummy indicating whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (left column), expected desired working hours over the next years (middle column, only if in labor force), or expected retirement age (right column, only if in labor force), all coded as 0 or 100. The underlying regressions are specifications 4, 5, and 6 in Table ??, which jointly include changes to retirement financial wealth, to other financial wealth, and to household net income. For each outcome (debt, desired working hours, and retirement age), we plot coefficients on changes in retirement wealth (top), changes in other financial wealth (middle), and by changes in household income (bottom) in percentage terms, respectively. All specifications control for all other changes to household income and/or wealth, trimmed at the 2nd and 98th percentiles, gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, log of total financial wealth, of real estate wealth, and of debt, share of total financial wealth in retirement accounts, borrowing constraints, stock market participation, stock shares in retirement and other accounts, investment experience, Census region, survey date, and the survey arm. All specifications are based on respondents in the four control arms, who have not received any information.

Figure A16: Expected duration of recovery across groups



Notes: This figure displays respondents' subjective expectations about the duration of the recovery in years for the US stock market, the respondent's pre-crisis household net wealth, and the respondent's pre-crisis household net income (left column) and the fractions of respondents who believe that their household net wealth or income will never recover (right column) by quintile of the pre-crisis net wealth distribution (top row), by quintile of the pre-crisis net household income distribution (middle row), and by age group (bottom row). The figures on expected recovery duration of the stock market and own wealth condition on those who have made financial losses, while the figures on income recovery duration condition on those who have incurred income losses. The sample consists of respondents in the pure control group, who have not received any questions or information on past crashes before answering to the questions on expected recovery duration.

B Additional tables

Table A1: Treatment and sample details

Treatment	Sample	Information	Respondents
Financial Crisis 2007	Information	5.5 years	1,055
	Control	-	1,050
Dot-com bubble	Information	7 years	1,041
	Control	-	1,063
Black Monday	Information	2 years	1,091
	Control	-	1,058
Pure control	Control	-	1,089
Total			7,447

Notes: The table gives an overview of the various control and treatment arms in the survey. The final number of participants is listed in the column *Respondents*.

Table A2: Integrity of treatment randomization

	FinCrisis Info	FinCrisis Control	DotCom Info	DotCom Control	Black Monday Info	Black Monday Control	Pure Control	P-value	Obs.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.51	0.50	0.52	0.51	0.51	0.52	0.53	0.815	7,447
Age	48.32	48.65	48.53	48.94	48.21	48.84	48.94	0.907	7,447
Bachelor's degree or higher	0.38	0.38	0.38	0.40	0.39	0.39	0.39	0.981	7,447
HH income (gross, USD)	80,074	80,836	82,281	81,628	81,753	79,662	80,472	0.939	7,417
Republican	0.37	0.37	0.39	0.36	0.36	0.36	0.37	0.817	7,447
Democrat	0.38	0.39	0.38	0.39	0.39	0.40	0.38	0.928	7,447
Region Midwest	0.25	0.24	0.25	0.23	0.25	0.27	0.23	0.470	7,447
- South	0.33	0.30	0.35	0.34	0.33	0.33	0.34	0.190	7,447
- West	0.21	0.25	0.21	0.22	0.20	0.21	0.21	0.273	7,447
Stock investor	0.61	0.64	0.61	0.62	0.61	0.59	0.61	0.565	7,447

Notes: The table shows respondent characteristics across the 7 treatment and control arms. Column 8 shows the p-Value of an F-test that all coefficients are zero when jointly regressing the respective characteristics on all treatment dummies.

Table A3: Average losses across groups and samples

Sort	Sample	Difference in losses between 5th and 1st quintile (%)		
		Other financial wealth	Retirement wealth	Total financial wealth
		(1)	(2)	(3)
Net wealth	Unconditional	-11.33	-15.70	-13.48
	Financial wealth > 0	-7.58	-10.23	-6.99
	Stocks > 0	-4.81	-3.85	-1.37
Net income	Unconditional	-10.50	-13.93	-12.07
	Financial wealth > 0	-8.18	-10.58	-8.23
	Stocks > 0	-5.23	-5.21	-3.23
Age group	Unconditional	0.21	-3.74	-2.67
	Financial wealth > 0	1.72	-3.28	-1.68
	Stocks > 0	2.79	-4.03	-1.52

Notes: The table shows the difference in percentage financial losses between the 5th and 1st quintile of the distributions of household net wealth and household net income, and between the lowest (18-24) and highest (65 and older) age categories. Differences in the percentage losses are displayed separately for non-retirement financial wealth (column 1), other financial wealth (column 2), and total financial wealth (column 3). We display differences between the two extreme quintiles (age categories) unconditionally for the entire sample, for the subsample of individuals with positive financial wealth holdings as of January 2020, and for the subsample of individuals with positive equity investments as of January 2020. Within each (sub-)sample, we include all respondents with nonmissing survey responses.

Table A4: Determinants of realized and planned adjustments to stock share: Pure control only

	Changed stock share	Increased stock share	Decreased stock share	Plan change stock share	Plan incr. stock share	Plan decr. stock share
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Net income (%)	-0.203* (0.106)	-0.022 (0.115)	-0.181 (0.113)	-0.267** (0.122)	0.029 (0.142)	-0.296*** (0.110)
Δ Retirement fin. wealth (%)				0.005 (0.181)	-0.203 (0.176)	0.209 (0.155)
Δ Other fin. wealth (%)				-0.480*** (0.175)	-0.348** (0.172)	-0.132 (0.160)
Ln(Total fin. wealth)	2.733 (1.740)	-1.487 (1.712)	4.220** (1.660)	0.960 (1.769)	0.109 (1.845)	0.851 (1.613)
Stock share in ret. wealth	-0.073 (0.069)	-0.020 (0.061)	-0.053 (0.055)	-0.106 (0.072)	-0.034 (0.063)	-0.071 (0.049)
Stock share in ot. fin wealth	-0.044 (0.065)	-0.015 (0.054)	-0.029 (0.058)	-0.036 (0.065)	-0.039 (0.056)	0.003 (0.053)
Share ret. in tot. fin. wealth	-0.074 (0.089)	-0.034 (0.086)	-0.040 (0.086)	-0.092 (0.093)	-0.063 (0.090)	-0.029 (0.081)
Any loss fin. crisis	4.355 (5.250)	0.934 (5.201)	3.421 (4.740)	-0.202 (5.343)	-4.541 (5.330)	4.339 (4.523)
Big loss fin. crisis	10.581* (5.805)	-1.029 (4.675)	11.611** (5.729)	-8.162 (5.398)	-10.969** (5.026)	2.807 (4.661)
Any loss dot-com	-6.620 (5.424)	-7.927 (5.038)	1.307 (4.961)	-2.290 (5.387)	5.642 (5.251)	-7.933** (3.952)
Big loss dot-com	-12.558 (7.925)	7.262 (6.896)	-19.821*** (6.592)	7.637 (8.543)	2.022 (8.043)	5.615 (6.503)
Any loss Black Monday	7.533 (6.558)	2.634 (5.830)	4.899 (5.917)	6.042 (6.147)	3.738 (5.808)	2.303 (4.492)
Big loss Black Monday	-11.810 (10.551)	-14.421** (6.596)	2.611 (9.362)	-12.090 (9.951)	-10.374 (8.497)	-1.716 (6.631)
Male	11.502** (4.970)	7.733* (4.494)	3.769 (4.297)	1.098 (4.944)	4.701 (4.594)	-3.603 (3.779)
At least bachelor	-22.711 (20.880)	-21.361 (22.314)	-1.350 (15.922)	10.944 (21.657)	-11.292 (25.271)	22.236* (12.633)
Republican	-6.972 (4.556)	-7.019 (4.352)	0.047 (4.268)	0.050 (4.445)	-0.710 (4.424)	0.760 (3.816)
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	.221	.135	.057	.279	.108	.102
Observations	550	550	550	503	503	503

Notes: This table shows OLS estimates of the determinants of realized and planned adjustments to the share of equities in total financial assets. The outcomes are dummies indicating whether the respondent's household has made any change, has increased or has decreased the equity share in total financial assets since the beginning of the stock market drop (columns 1-3) and dummies indicating plans to change, increase or decrease the equity share in overall financial assets in the weeks after the survey (columns 4-6), all coded as 0 or 100. All specifications are based on the pure control group, which has not received any information and not answered any questions on past crashes, using only respondents who report positive stockholdings as of January 2020. All specifications control for shocks to income, trimmed at the 2nd and 98th percentiles, dummies for having lost any wealth or substantial wealth during past stock market crashes, gender, age, employment status, being the main earner, being the financial decision-maker, party affiliation, log net household income, log of total financial wealth, of real estate wealth, and of debt, share of total financial wealth in retirement accounts, borrowing constraints, stock market participation, stock shares in retirement and other accounts, investment experience, Census region, survey date, and the survey arm. Columns 4-6 additionally control for shocks to retirement and other financial wealth, trimmed at the 2nd and 98th percentiles. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A5: Income and wealth shocks and expected economic decisions: Pure control only

	Exp. spend. growth (%)	Exp. spend. growth (\$)	Exp. spend. growth (\$)	Incr. exp. debt	Incr. exp. desired hours	Incr. exp. retirement age
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Retirement fin. wealth (%)	0.018 (0.068)			-0.148 (0.148)	-0.274 (0.188)	-0.696*** (0.195)
Δ Other fin. wealth (%)	-0.061 (0.070)			-0.443*** (0.140)	-0.371** (0.176)	0.006 (0.179)
Δ Net income (quarterly, %)	0.189*** (0.048)			-0.286*** (0.085)	-0.358*** (0.105)	-0.401*** (0.100)
Δ Retirement fin. wealth (\$)		0.048*** (0.016)	0.039** (0.017)			
Δ Other fin. wealth (\$)		0.011 (0.019)	-0.002 (0.021)			
Δ Net income (quarterly, \$)		0.584*** (0.201)				
Δ Net income (annual, \$)			0.536*** (0.139)			
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	.048	.044	.114	.198	.108	.124
Observations	970	917	906	987	603	603

Notes: This table shows estimates of the association of shocks to the respondent's household net income and financial wealth with expected economic decisions. The outcomes are expected growth of yearly household spending from 2019 to 2020 in percent, trimmed at the 2nd and 98th percentiles (column 1); expected household spending growth in dollars, trimmed at the 2nd and 98th percentiles (columns 2-3); and dummies indicating whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (column 4), expected desired working hours over the next years (column 5, only if in labor force) or expected retirement age (column 6, only if in labor force), coded as 0 or 100. Dollar changes in columns 2 and 3 are constructed from survey questions for retirement and other financial wealth and for income (assuming that the respondent expected a quarter of her 2019 income in the first quarter of 2020), and for spending from the survey question on percent changes and estimates of the level of spending of different groups in 2019 from the CEX. Columns 1, 2, 4, 5 and 6 show simple OLS estimates. Column 3 shows a 2SLS estimate, where the respondent's expected dollar change in household income from 2019 to 2020 is instrumented with the unexpected shock to household income in the first quarter. All specifications are based on the pure control group, which has not received any information and not answered any questions on past crashes. All specifications control for shocks to income and financial wealth, trimmed at the 2nd and 98th percentiles, gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, risky portfolio share, investment experience, Census region, survey date, and the survey arm. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A6: Determinants of expectations about the stock market and own wealth:
Pure control only

	Stock recovery duration	Stock return: Mean	Stock return: SD	Stock return <-30%	Stock return >30%	Wealth recovery duration	Exp. wealth never to recover	Household financial prospects
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Fin. wealth (%)	-0.010* (0.006)	0.020 (0.058)	-0.024 (0.026)	-0.032 (0.065)	-0.041 (0.086)	-0.022*** (0.005)	-0.229** (0.113)	0.010*** (0.003)
Δ Net income (%)	-0.002 (0.004)	0.087*** (0.033)	-0.005 (0.018)	-0.111** (0.043)	0.087* (0.044)	-0.006** (0.003)	-0.065 (0.067)	0.004** (0.002)
Any loss fin. crisis	-0.138 (0.156)	-1.868 (1.643)	0.234 (0.803)	-1.043 (1.798)	-3.853 (2.417)	0.057 (0.124)	4.437 (3.473)	-0.113 (0.087)
Big loss fin. crisis	0.168 (0.215)	0.693 (2.222)	-0.119 (0.990)	-0.744 (2.414)	3.531 (3.386)	-0.043 (0.170)	2.799 (4.205)	-0.098 (0.111)
Any loss dot-com	0.100 (0.171)	-1.673 (1.776)	0.493 (0.890)	3.121 (2.180)	-0.379 (2.549)	0.229 (0.161)	-3.492 (3.446)	-0.046 (0.097)
Big loss dot-com	-0.537** (0.234)	-0.211 (3.496)	-1.509 (1.497)	-1.645 (3.785)	2.219 (5.506)	-0.302 (0.242)	-3.943 (6.226)	0.340* (0.178)
Any loss Black Monday	0.139 (0.215)	-3.562 (2.268)	0.123 (1.014)	0.860 (2.789)	-6.867** (3.124)	0.138 (0.191)	-3.158 (4.107)	-0.010 (0.117)
Big loss Black Monday	-0.118 (0.254)	0.864 (3.843)	-0.200 (1.496)	0.595 (4.885)	2.592 (4.876)	-0.206 (0.237)	21.798** (8.856)	-0.246 (0.195)
Male	-0.170 (0.169)	-0.267 (1.418)	1.488** (0.682)	-0.556 (1.716)	0.515 (2.033)	-0.107 (0.093)	4.130 (2.803)	0.279*** (0.076)
At least bachelor	-0.258 (0.580)	0.156 (4.319)	-1.612 (1.874)	-9.319 (6.354)	-4.785 (5.214)	-0.169 (0.241)	-0.711 (7.900)	0.194 (0.182)
Republican	-0.194 (0.151)	6.885*** (1.430)	-0.271 (0.665)	-4.814*** (1.733)	7.546*** (2.113)	-0.249** (0.108)	-4.913* (2.780)	0.325*** (0.072)
Stock investor	-0.494* (0.279)	-1.219 (2.512)	2.124 (1.341)	2.196 (3.233)	-1.204 (3.619)	0.613*** (0.210)	-0.679 (5.395)	0.165 (0.144)
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	.045	.049	.049	.017	.055	.234	.075	.091
Observations	1,014	1,014	1,014	1,014	1,014	941	1,014	1,014

Notes: This table shows OLS estimates of the determinants of respondents' expectations about the stock market and their own wealth. The outcomes are the expected duration of the recovery from the current crash in years (column 1); mean and standard deviation as well as probabilities assigned to extreme return realizations based on the respondent's reported probability distribution over the one year-ahead stock market return (columns 2-5); the expected recovery duration of the respondent's household financial wealth (column 6); a dummy indicating whether the respondent thinks her household net wealth will never recover, coded as 0 or 100 (column 7); and a categorical measure of the respondent's subjective household financial prospects, z-scored using the mean and standard deviation in the sample (column 8). All specifications are based on the pure control group, which has not received any information and not answered any questions on past crashes. All specifications control for shocks to income and financial wealth, trimmed at the 2nd and 98th percentiles, dummies for having lost any wealth or substantial wealth during past stock market crashes, gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, risky portfolio share, investment experience, Census region, and survey date. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A7: Correlational evidence on beliefs about past crashes

	Stock recovery duration	Agree: Recovery many yrs.	Agree: Set back many yrs.	Agree: Recover 2020	Stock return: Mean	Stock return: SD	Stock return <-30%	Stock return >30%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Fin Crisis								
Prior recovery duration Fin. Crisis 2007	0.245*** (0.029)	0.049*** (0.009)	0.048*** (0.009)	-0.017* (0.010)	-0.153 (0.169)	0.101 (0.094)	0.153 (0.231)	-0.159 (0.252)
Observations	1,050	1,050	1,050	1,050	1,050	1,050	1,050	1,050
Panel B: Dot-com								
Prior recovery duration Dot-com 2000	0.225*** (0.025)	0.032*** (0.008)	0.031*** (0.008)	-0.010 (0.008)	0.159 (0.153)	0.135* (0.076)	0.010 (0.196)	0.355 (0.228)
Observations	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063
Panel C: Black Monday								
Prior recovery duration Black Monday 1987	0.313*** (0.026)	0.058*** (0.007)	0.051*** (0.008)	-0.034*** (0.008)	-0.409*** (0.153)	0.229*** (0.073)	0.541*** (0.201)	-0.217 (0.228)
Observations	1,058	1,058	1,058	1,058	1,058	1,058	1,058	1,058
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS estimates of the effect of prior beliefs about the duration of a past stock market crash on the respondent's expectations about the stock market. The outcomes are the expected duration of the recovery from the current crash in years (column 1); agreement on 7-point scales to statements describing the severity of the current stock market crash, z-scored using the mean and the standard deviation in the sample (columns 2-4); mean and standard deviation as well as probabilities assigned to extreme return realizations based on the respondent's reported probability distribution over the one year-ahead stock market return (columns 5-8). Panel A is based on the control arm including questions on the Financial Crisis 2007. Panel B is based on the control arm including questions on the burst of the Dot-com bubble 2000. Panel C is based on the control arm including questions on the Black Monday 1987. All specifications control for gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in total financial assets, investment experience, Census region and survey date. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A8: Effects of information on own outlook and plans: Experimental reduced form

	Wealth recovery duration	Household financial prospects	Plan incr. stock share	Plan decr. stock share	Exp. spend. growth	Incr. exp. debt	Incr. exp. desired hours	Incr. exp. retirement age
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: All								
Info Fin. Crisis 2007	0.484*** (0.078)	-0.100** (0.043)	0.952 (1.931)	3.054* (1.818)	-0.951 (0.914)	0.335 (1.924)	6.429** (2.632)	5.186* (2.662)
Panel B: Underestimators								
Info Fin. Crisis 2007	0.522*** (0.086)	-0.099* (0.052)	2.827 (2.333)	3.215 (2.211)	-1.478 (1.106)	2.978 (2.381)	7.401** (3.210)	8.229** (3.235)
Panel C: Under. & Stocks > 0								
Info Fin. Crisis 2007	0.761*** (0.130)	-0.136** (0.066)	2.827 (2.333)	3.215 (2.211)	-1.281 (1.328)	3.101 (3.002)	9.099** (3.773)	8.995** (3.917)
Panel D: All								
Info Dot-com 2000	0.573*** (0.087)	0.009 (0.043)	0.285 (2.015)	-2.085 (1.843)	-1.255 (0.928)	-1.477 (1.909)	-4.891* (2.623)	-3.710 (2.664)
Panel E: Underestimators								
Info Dot-com 2000	0.683*** (0.109)	0.022 (0.054)	1.149 (2.601)	-2.967 (2.279)	-0.533 (1.141)	-1.019 (2.334)	-4.145 (3.251)	-3.243 (3.321)
Panel F: Under. & Stocks > 0								
Info Dot-com 2000	1.017*** (0.164)	0.051 (0.069)	1.149 (2.601)	-2.967 (2.279)	0.779 (1.374)	0.167 (2.898)	-1.328 (3.822)	-3.998 (3.974)
Panel G: All								
Info Black Monday 1987	-0.157** (0.074)	0.048 (0.042)	3.293* (1.984)	-2.228 (1.710)	-1.027 (0.924)	-5.598*** (1.877)	-0.566 (2.612)	-0.806 (2.611)
Panel H: Overestimators								
Info Black Monday 1987	-0.242*** (0.088)	0.088* (0.047)	4.647** (2.253)	-4.079** (1.981)	-0.066 (1.070)	-6.705*** (2.141)	-1.607 (2.953)	-4.698 (2.962)
Panel I: Over. & Stocks > 0								
Info Black Monday 1987	-0.385*** (0.146)	0.097 (0.062)	4.647** (2.253)	-4.079** (1.981)	1.296 (1.361)	-5.026* (2.709)	-1.170 (3.558)	-3.673 (3.609)
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS estimates of the effect of being shown information on the duration of a past stock market crash on the respondent's expectations about her own financial situation and behavior. The outcomes are the expected recovery duration of the respondent's household financial wealth (column 1); a categorical measure of the respondent's subjective household financial prospects, z-scored using the mean and standard deviation in the sample (column 2); dummies indicating plans to increase or decrease the risky share in overall financial assets in the weeks after the survey (columns 3-4, only for stockholders); expected growth of yearly household spending from 2019 to 2020 in percent, trimmed at the 2nd and 98th percentiles (column 5); dummies indicating whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (column 6), expected desired working hours over the next years (column 7, only if in labor force) or expected retirement age (column 8, only if in labor force). All dummy outcomes are coded as 0 or 100. Panels A-C are based on the treatment and control arms including information or questions on the Financial Crisis 2007. Panels D-F are based on the treatment and control arms including information or questions on the burst of the Dot-com bubble 2000. Panels G-I are based on the treatment and control arms including information or questions on the Black Monday 1987. Panels A, D and G are based on the full sample in the corresponding arms. Panels B, E and H are based only on under-estimators (for Financial Crisis and Dot-com bubble) or over-estimators (for Black Monday) of the length of the recovery from the crash. Panels C, F and I are based only on under-estimators or over-estimators who participated in the stock market before the current crisis. All specifications control for the respondent's prior belief about the recovery duration following the corresponding crash as well as for gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in total financial assets, investment experience, Census region and survey date. The specifications on planned stock trading in columns 3 and 4 also control for realized trading since the onset of the crisis. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A9: Effects of expected stock market recovery on own outlook and plans: Heterogeneity

	Wealth recovery duration	Household financial prospects	Plan incr. stock share	Plan decr. stock share	Exp. spend. growth	Incr. exp. debt	Incr. exp. desired hours	Incr. exp. retirement age
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Age $\leq$ median								
Expected stock recovery duration (years)	0.244*** (0.078)	-0.047*** (0.012)	-0.809* (0.420)	0.396 (0.415)	-0.317 (0.272)	-0.145 (0.490)	2.085*** (0.525)	1.922*** (0.535)
Panel B: Age $>$ median								
Expected stock recovery duration (years)	0.995*** (0.082)	-0.128*** (0.011)	-0.190 (0.356)	0.750** (0.375)	-0.730*** (0.222)	1.236** (0.496)	2.179** (0.854)	2.136** (0.909)
p-value (A=B)	0.00	0.00	0.26	0.53	0.24	0.05	0.93	0.84
Panel C: Net income $\leq$ median								
Expected stock recovery duration (years)	0.529*** (0.099)	-0.085*** (0.012)	-0.739* (0.402)	0.506 (0.440)	-0.348 (0.292)	1.234** (0.534)	2.063*** (0.744)	1.407* (0.790)
Panel D: Net income $>$ median								
Expected stock recovery duration (years)	0.698*** (0.070)	-0.090*** (0.012)	-0.554 (0.398)	0.580 (0.356)	-0.762*** (0.226)	0.158 (0.431)	2.184*** (0.556)	2.853*** (0.570)
p-value (C=D)	0.16	0.72	0.74	0.90	0.26	0.12	0.90	0.14
Panel E: Net wealth $\leq$ median								
Expected stock recovery duration (years)	0.739*** (0.066)	-0.102*** (0.010)	-0.540 (0.336)	0.669** (0.312)	-0.649*** (0.204)	0.695* (0.385)	2.421*** (0.553)	2.661*** (0.573)
Panel F: Net wealth $>$ median								
Expected stock recovery duration (years)	0.348*** (0.116)	-0.057*** (0.015)	-0.709 (0.506)	0.168 (0.552)	-0.332 (0.343)	0.253 (0.657)	1.289 (0.791)	1.107 (0.788)
p-value (E=F)	0.00	0.02	0.78	0.43	0.43	0.56	0.24	0.11
Panel G: Female								
Expected stock recovery duration (years)	0.488*** (0.082)	-0.078*** (0.011)	-0.750** (0.355)	0.570 (0.380)	-0.586** (0.230)	0.339 (0.477)	2.755*** (0.689)	1.871*** (0.682)
Panel H: Male								
Expected stock recovery duration (years)	0.710*** (0.084)	-0.102*** (0.013)	-0.409 (0.456)	0.516 (0.420)	-0.575** (0.277)	0.669 (0.482)	1.508** (0.595)	2.211*** (0.631)
p-value (G=H)	0.06	0.16	0.55	0.92	0.98	0.63	0.17	0.71
Panel I: Below Bachelor								
Expected stock recovery duration (years)	0.507*** (0.085)	-0.083*** (0.011)	-0.596 (0.364)	0.673* (0.364)	-0.359 (0.241)	0.986** (0.461)	2.030*** (0.589)	2.093*** (0.620)
Panel J: At least Bachelor								
Expected stock recovery duration (years)	0.714*** (0.078)	-0.092*** (0.013)	-0.694 (0.436)	0.497 (0.409)	-0.730*** (0.256)	-0.034 (0.496)	2.277*** (0.683)	2.082*** (0.679)
p-value (I=J)	0.07	0.61	0.86	0.75	0.29	0.13	0.78	0.99
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS estimates of the effects of respondents' expected stock market recovery duration on the respondent's expectations about her own situation and behavior for different subgroups. The outcomes are the expected recovery duration of the respondent's household net wealth (column 1); a categorical measure of the respondent's subjective household financial prospects, z-scored using the mean and standard deviation in the sample (column 2); dummies indicating plans to increase or decrease the risky share in overall financial assets in the weeks after the survey (columns 3-4, only for stockholders); expected growth of yearly household spending from 2019 to 2020 in percent, trimmed at the 2nd and 98th percentiles (column 5); dummies indicating whether the coronavirus crisis increases the respondent's expectations about outstanding household debt by the end of 2020 (column 6), expected desired working hours over the next years (column 7, only if in labor force) or expected retirement age (column 8, only if in labor force). All dummy outcomes are coded as 0 or 100. All estimations are based on the four control arms, which have not received any information, and are restricted to those who participated in the stock market before the current crisis. All specifications control for gender, age, employment status, being the main earner, being financial decision-maker, party affiliation, log net household income, logs of retirement wealth, of other financial wealth, of real estate wealth, and of debt, borrowing constraints, stock market participation, the equity share in total financial assets, investment experience, Census region and survey date. The specifications on planned stock trading in columns 3 and 4 also control for realized trading since the onset of the crisis. Robust standard errors are reported in parentheses. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.

Table A10: Perceived purpose of the survey

	All	All, excl. pure ctrl	Info treat., all	Ctrl. all, ex pure ctrl.	Difference (4) – (3)
	(1)	(2)	(3)	(4)	(5)
Impact COVID-19 on HH finances	47.25	46.29	44.62	47.97	3.35** [2.68]
Expectations	10.23	10.54	11.39	9.68	-1.71* [-2.22]
Knowledge test	2.73	2.88	3.36	2.40	-0.96* [-2.29]
Comparison of fin. crises	2.32	2.56	3.01	2.11	-0.90* [-2.27]
Experiment	0.07	0.08	0.09	0.06	-0.03 [-0.44]
Do not know	3.32	3.40	3.83	2.96	-0.86 [-1.90]
Other	34.08	34.26	33.70	34.82	1.12 [0.94]
Observations	7,447	6,358	3,187	3,171	

Notes: This table shows the relative frequency (in %) of answers to the question “What do you think was the purpose of the survey?” Answers were given as free text entries. We manually categorize the answers into 8 categories based on meaningful keywords, (1) impact of the corona crisis on household finances, (2) household economic expectations, (3) knowledge test/education, (4) comparison of different financial crises, (6) some form of experiment, (7) do not know, and (8) other. Column 1 is based on the entire sample. Column 2 excludes respondents in the pure control group, who have not received any questions or information on previous crashes. Column 3 is based on respondents in the three information treatment arms *FinCrisisInfo*, *DotComInfo* and *BlackMondayInfo*. Column 4 is based on respondents in the three control treatment arms *FinCrisisCtrl*, *DotComCtrl* and *BlackMondayCtrl*, excluding the pure control group. Column 5 shows the differences across percentages in the treatment and control arms, excluding the pure control group. * denotes significance at the 10 pct., ** at the 5 pct., and *** at the 1 pct. level.